Looking towards the future

Big questions

DevSecOps is constantly growing:

- People learn new skills.
- Process change to meet demand.
- Technology adapts to delivery.

Flow and feedback in small intervals helps ensure pace.

Staying on the leading edge means continuous learning:

- Encourage people.
- Adapt processes with experimentation.
- Try new technologies.

Find the best way to spend part of the pipeline on reaching the third horizon.

What most intrigues your team?



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Path forward

Gartner emerging Tech Roadmap 2020-2022

- Gartner regularly forecasts technology and trends.
- Finds market leaders and up-and-comers.

Technology says emerging, already tested, and likely in market

- Size of circle is enterprise value.
- Color is deployment risk.



Source: https://emtemp.gcom.cloud/ngw/globalassets/en/information-technology/documents/benchmarks/emerging-tech-roadmap-mse-2020-2022.pdf

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AI/ML (1/3)

AI is intelligence demonstrated by machines:

- can be general (Robby the Robot) or specific intelligence (Deep Blue, chess)
- offers broad applications
- changes with generations.

ML is the study of computer algorithms that can improve automatically through experience and by the use of data.

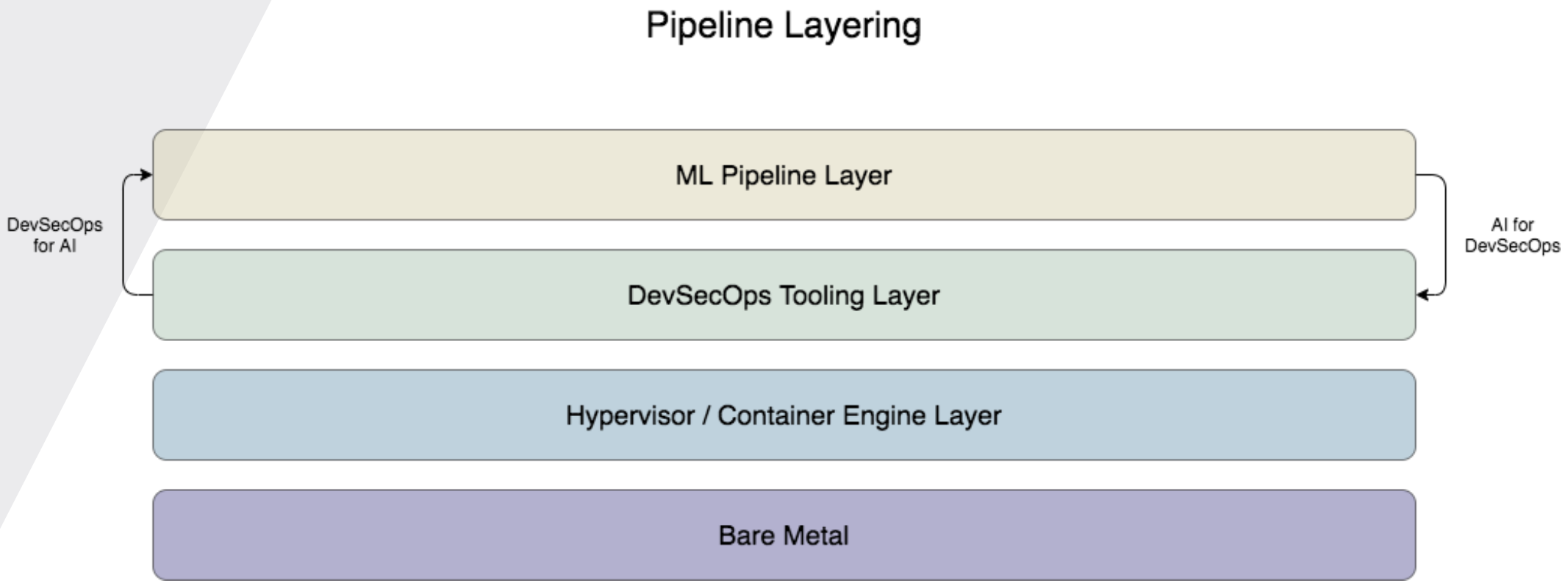
- relies on algorithm-based processes
- lacks robust solutions
- evaluates inputted data.

Layered pipelines with ML and DevOps approaches



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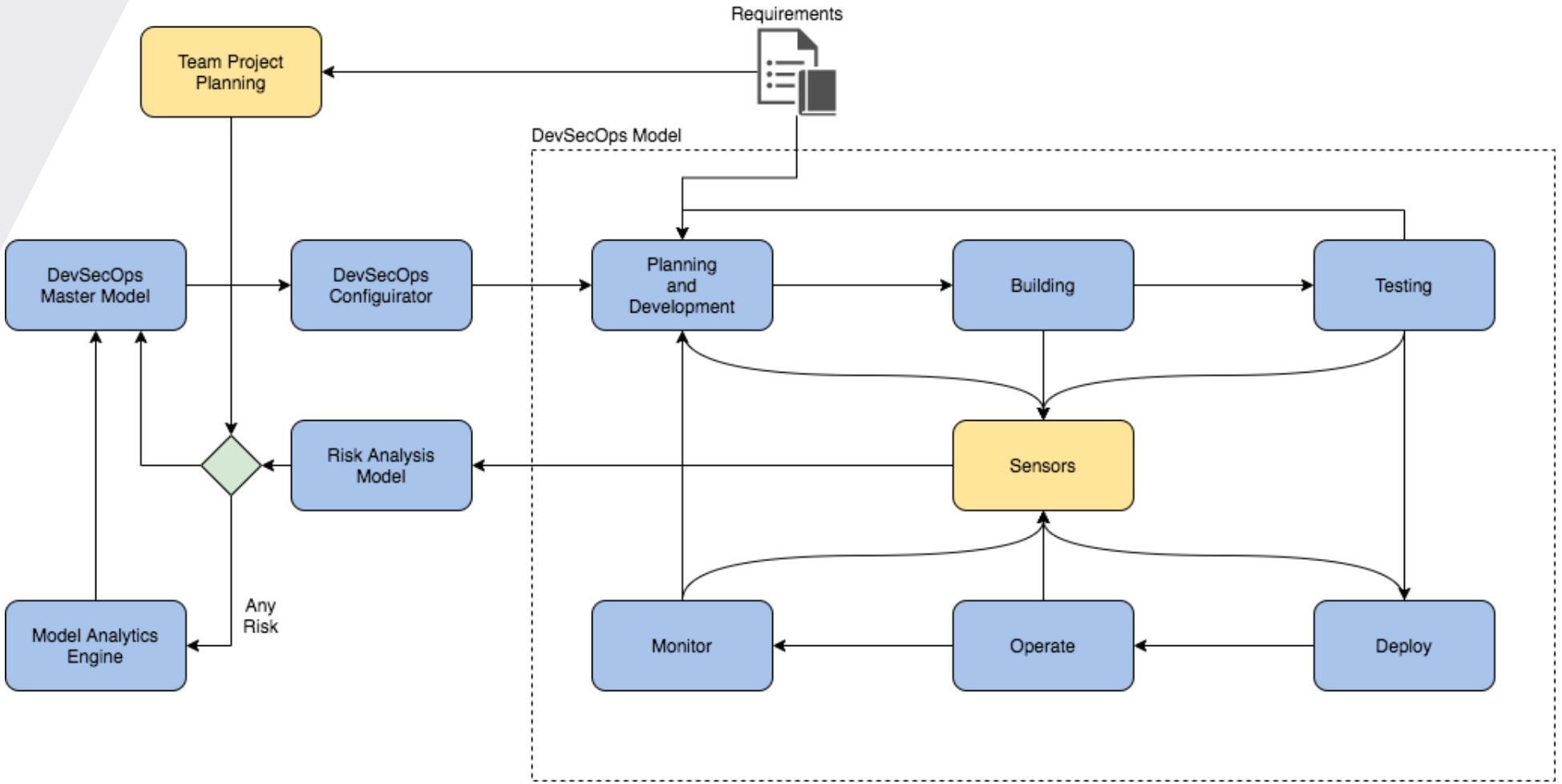
AI/ML (2/3)



Source: https://insights.sei.cmu.edu/blog/aligning-devsecops-and-machine-learning/

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AI/ML (3/3)



Source: https://insights.sei.cmu.edu/blog/aligning-devsecops-and-machine-learning/

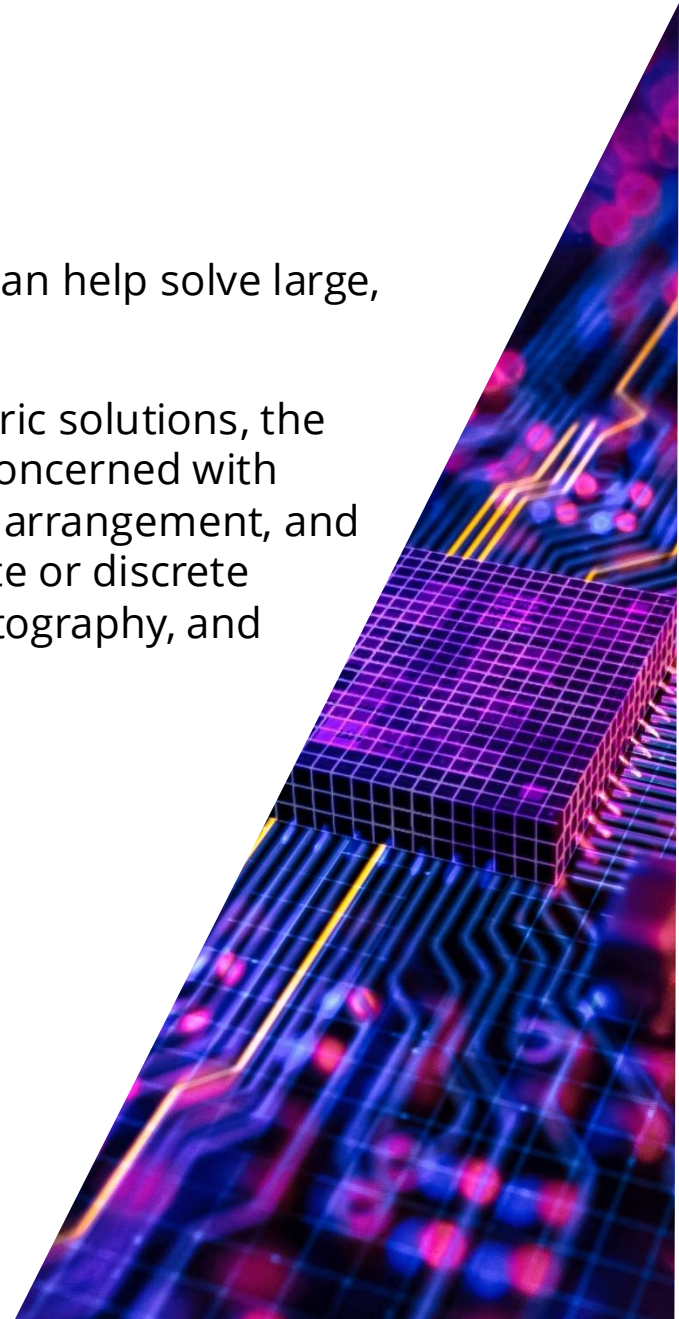
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Quantum computing

Quantum computing harnesses the collective properties of quantum states, such as superposition, interference, and entanglement, to perform calculations.

- **Superposition:** any two (or more) quantum states can be added together (superposed), and the result will be another valid quantum state.
- **Interference:** each quantum particle interferes with itself in a wave state and which possibilities are finally assumed in a measurement on an initial superposition of options and states is entirely random.
- **Entanglement:** the phenomenon whereby a pair of particles are generated in such a way that the individual quantum states of each are indefinite until measured, and the act of measuring one determines the result of measuring the other, even when at a distance from each other.

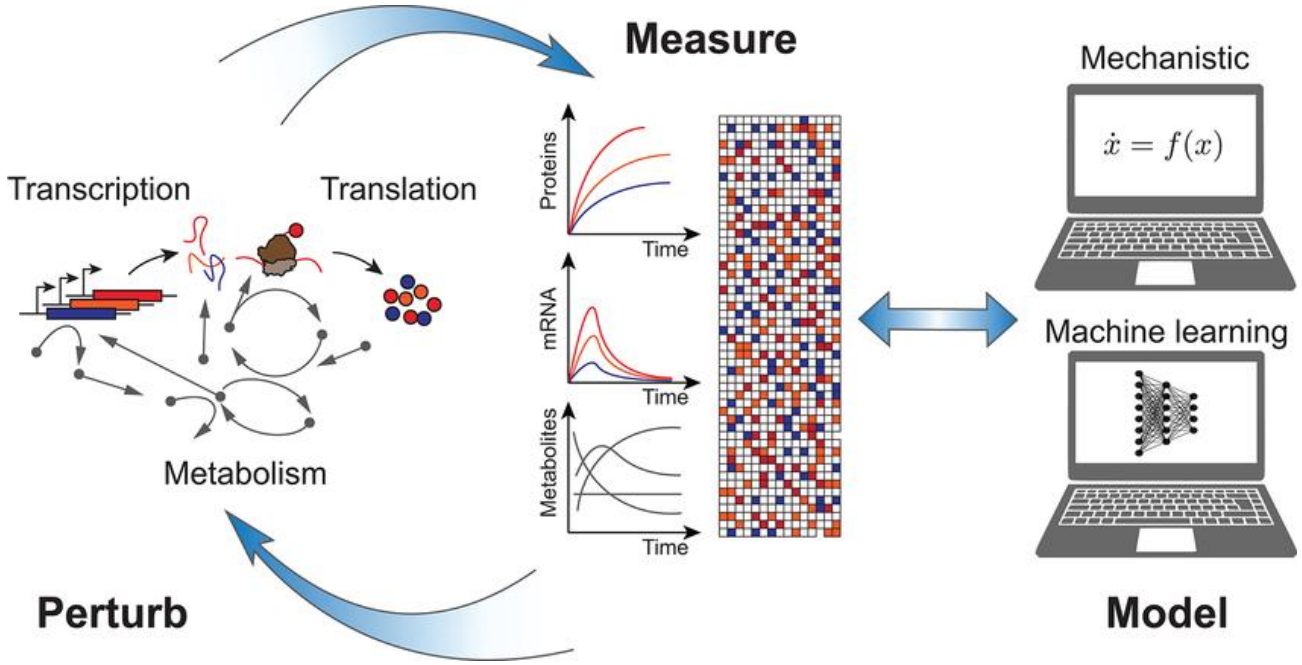
- Quantum computing can help solve large, complex problems.
- Works with combinatoric solutions, the field of mathematics concerned with problems of selection, arrangement, and operation within a finite or discrete system (weather, cryptography, and so on).



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Biodesign

- **Biodesign is the art of using living materials computers:**
  - cells rather than chips
  - protein switches rather than electrical.
- **Expands range of possibility:**
  - changes in power
  - rapid computing expanse
  - think using DNA to perform calculations.
- **Neural networks with nerves.**
- **Advantages:**
  - speed
  - power usages
  - complexity.
- **Multiple areas studying a variety of applications.**
- **Theoretically uses ML to understand protein applications.**



Source: https://www.researchgate.net/figure/Rational-computer-aided-biodesign-with-cell-free-systems-cell-free-gene-expression_fig1_343194953

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On-demand hardware

Hardware when needed	More than the IaaS, SaaS, and PaaS debate
	Most resembles IaaS solutions
	Mimics some basic cloud properties
Expands through specific hardware opportunities for scaled systems	Popular to support SCADA systems, legacy applications, and transitional environments.
	Sometimes specific hardware is needed.
	Offers virtual modelling to unique types.



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